

Model Card: BBO Gaussian Process Optimizer

Overview

- **Name:** Scaled Matern-UCB Optimizer
- **Type:** Gaussian Process Regression (GPR) with Upper Confidence Bound (UCB) Acquisition.
- **Version:** 2.0 (Enhanced Regularisation & Scaling)

Intended Use

- **Suitable for:** 2D, 3D, 4D, 5D, 6D & 8D Black-Box Optimisation (BBO) where function evaluations are expensive. It is ideal for continuous search spaces with moderate noise.
- **Avoid:** Discrete/categorical input spaces or functions with extreme discontinuities. Not intended for big data regimes due to the $O(n^3)$ complexity of GP fitting.

Details (Strategic Evolution)

The strategy evolved from a standard GP to a robust, noise-aware surrogate model. Across the competition rounds, the approach shifted to focus on three pillars:

1. **Preprocessing:** Implementing StandardScaler for both X and y to ensure the GP kernel operates on a uniform isotropic scale.
2. **Surrogate Refinement:** Using a Matern kernel ($\nu=2.5$) for realistic smoothness and increasing alpha to 0.01 to handle noise and prevent numerical instability.
3. **Exploration Strategy:** Employing **Latin Hypercube Sampling (LHS)** to generate 30,000 candidates in 8D space, ensuring the UCB function ($\kappa=2.0$) has a dense pool of points to evaluate for the next best sample.

Performance

- **Metrics:** Training R^2 , Prediction Uncertainty (σ), and UCB Score.
- **Summary:** The model achieves high R^2 scores, but performance is monitored via "overfit checks" ($R^2 > 0.99$). By unscaling predictions back to the original units, the model provides physically meaningful estimates of the global maximum across all eight test functions.

Assumptions and Limitations

- **Assumptions:** Assumes the underlying function is relatively smooth

(Stationarity) and that noise is normally distributed.

- **Limitations:** The "curse of dimensionality" in 8D means that even with 30,000 candidates, large "empty" regions of the search space remain.
- **Failure Modes:** If the true function has sharp peaks between sampled points, the Matern kernel may over-smooth and miss the global optimum.

Ethical Considerations

Transparency in this model card ensures **reproducibility**. By documenting the exact kernel parameters and scaling steps, other researchers can adapt this for real-world safety-critical engineering (e.g., experimental design). Clear documentation prevents the "black box" effect, allowing users to trust the uncertainty bounds provided by the GP.

Reflection on Decision-Making

How the approach makes decisions:

The model makes decisions by balancing **exploitation** (predicted high y) and **exploration** (high uncertainty σ). By using `np.argmax(ucb)`, it mathematically chooses the point with the highest potential "ceiling" rather than just the highest current average.

Strengths and Limitations:

The primary strength is **robustness**; manual scaling and noise regularisation (α) prevent the model from crashing on "spiky" data. The limitation is **computational density**; searching an 8-dimensional cube effectively requires significantly more points than 30,000, which can lead to missed local optima.

Value of Detail:

Adding more detail (like a version history of specific κ values used in early vs. late rounds) would improve clarity. It would help identify if the model became too "greedy" (exploitative) too early in the process. However, the current structure is sufficient for technical hand-off as it covers the critical mathematical transformations required to replicate the results.